Settings

Pollutant

- ○ PM10
- ◉ NO2

Date range

2019 / 01 / 23 – 2020 / 01 / 31

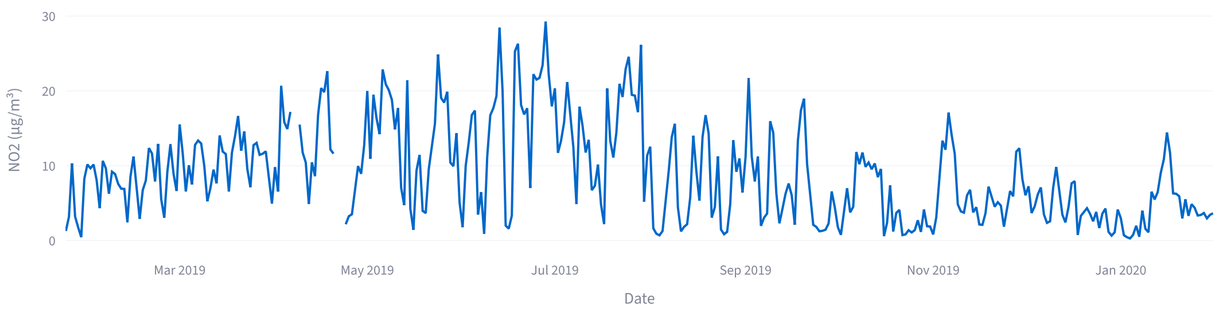
Henderson Air Quality Dashboard

Hourly NO2 and weather at Henderson monitoring site · 2019-01-23 to 2020-01-31

Average NO2	Highest hour	Average wind speed	Hours with data
8.9 µg/m³	76.6 µg/m³	1.5 m/s	98%

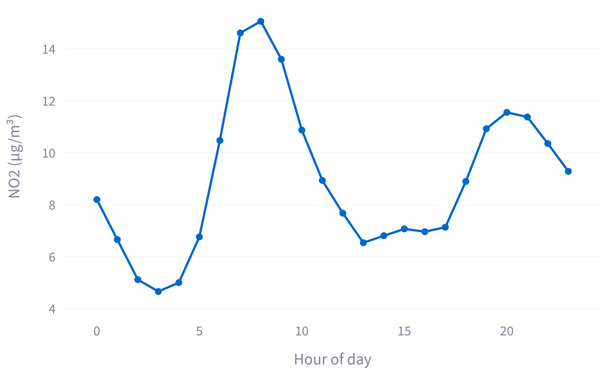
NO2 over time

Daily average

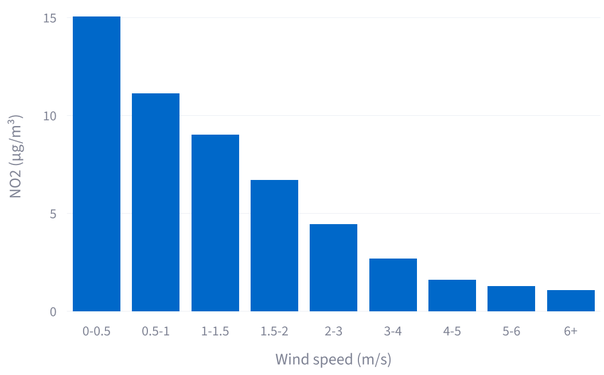


Patterns

Average by hour of day

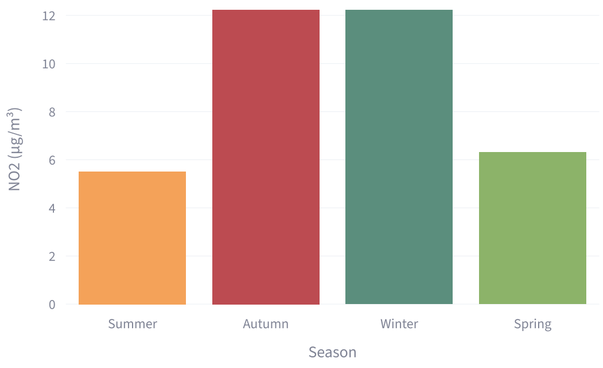


Average by wind speed

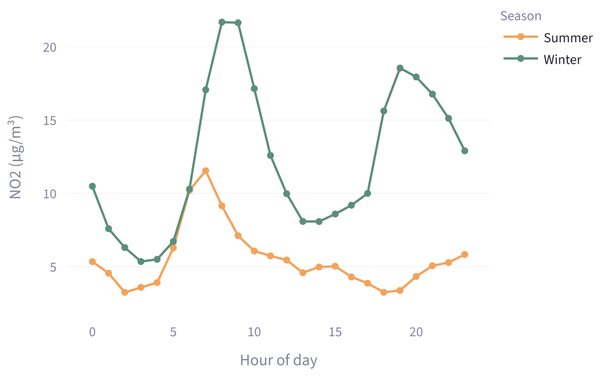


Seasons

Average by season



Summer vs winter: average by hour of day

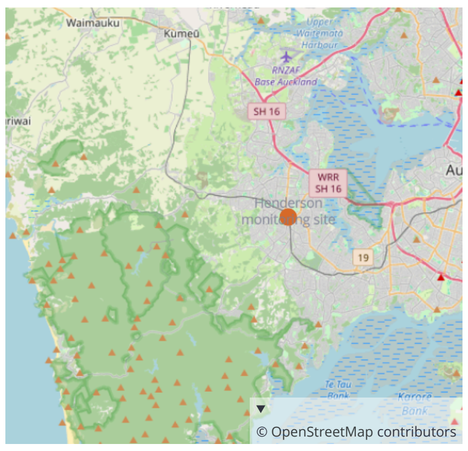


Summer average	Winter average	Winter vs summer
5.5 µg/m³	12.2 µg/m³	+122%

Wind direction and pollution

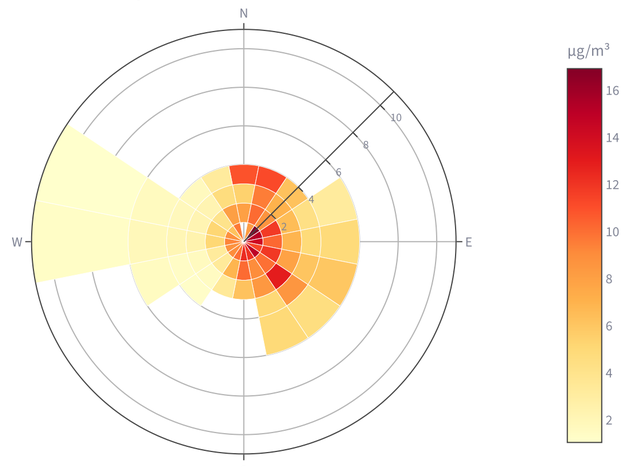
Wind direction was only recorded from September 2022 to February 2023, so these plots always use that period. Angle = direction the wind blows from (north at the top), distance from centre = wind speed, colour = average concentration.

Monitoring site



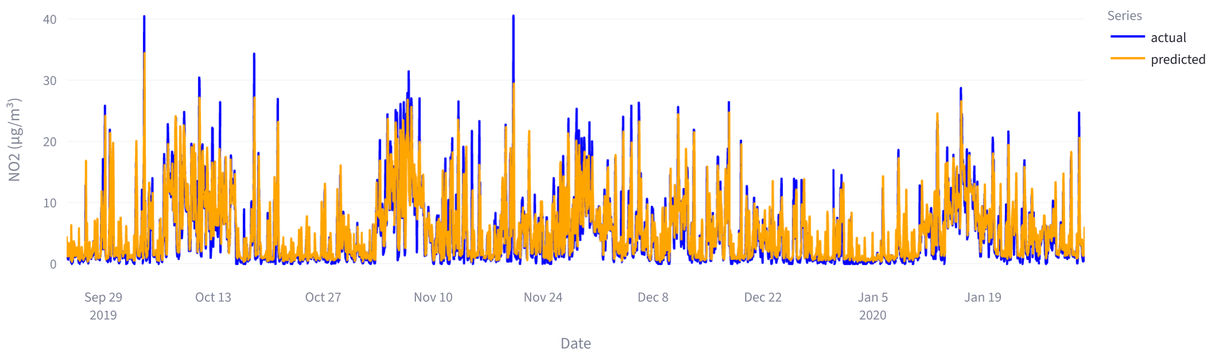
Henderson, Auckland · 3,654 hours with wind direction

NO_2 by wind direction and speed



Forecast: actual vs predicted

Model	MAE	RMSE	R^2
LightGBM	1.99	3.14	0.688



These metrics are recalculated for the selected dates, not the full held-out test set. Zoom in to inspect hourly forecasts.

Historical what-if NO2 forecast

Historical simulation only — not a real-time monitoring feed. Forecasts below use locally saved, validation-selected models. The preceding charts display published Supabase test-set predictions.

Forecast date

2023 / 01 / 25

Forecast origin hour

12:00

Forecast: next hour

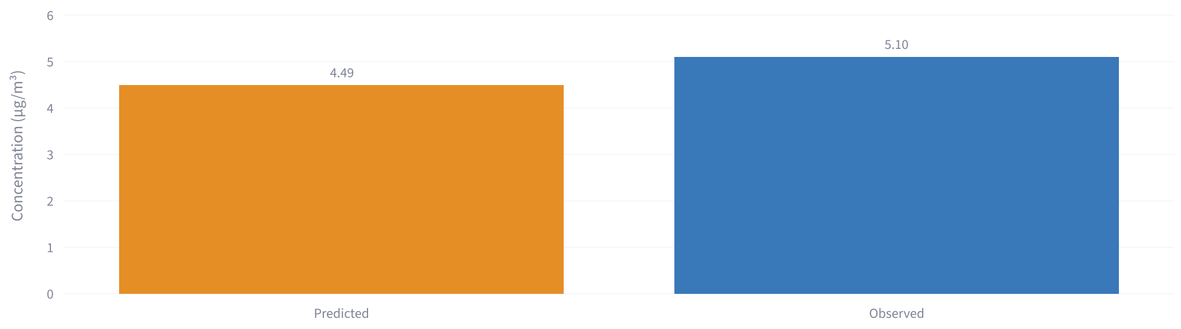
› What if the weather were different? (optional)

Predicted NO2 (1h)

4.49 µg/m³

Observed following 1h: 5.10 µg/m³

NO2: Actual vs Predicted (1-hour forecast)



Comparison for the selected historical forecast only; the graph is not an overall model performance measure.

Saved model: LightGBM · 14 input features

Henderson historical measurements · Supabase REST API (read-only). Interactive forecasts use historical stored inputs, which should be checked against the final forward-only preprocessing rules before claiming exact reproduction of the notebook's held-out scores. Monitoring site map location is approximate.